

Video RAG Retrieval Project

<https://github.com/aiyshwary/Video-RAG-Retrieval>

OVERVIEW

Python **FAISS** **OpenCLIP** **ViT-GPT2** **Sentence Transformers** **RAG**
A semantic video search system that extracts frames, generates scene captions via ViT-GPT2, and embeds

IMPACT

Enables semantic search over unstructured video content by leveraging vision-language models and vec

TECHNICAL WALKTHROUGH

Project Overview

Implements a simple Video Retrieval-Augmented Generation (RAG) system for semantic video search

Features

- i) Extract frames and scene descriptions from videos using ViT-GPT2 image captioning.
- ii) Compute text embeddings with SentenceTransformers and image embeddings with OpenCLIP.
- iii) Store embeddings in FAISS indexes with associated metadata (video name, scene ID, frame number).
- iv) Retrieve relevant scenes or frames for a text query using cosine similarity.

Installation

Clone the repository, create a virtual environment, and install dependencies from requirements.txt. GPU

Example Usage

Process a video and index its scenes: `extract_scenes('path/to/video.mp4', fps=0.5); retriever = VideoRAG`
Query: `results = retriever.query('a person walking', k=3, use_text=True)`.

Project Layout

/Video RAG Retrieval Project ■■■■ README.md ■■■■ requirements.txt ■■■■ src/video_rag/ ■■■■

Notes

i) Scene detection is rudimentary; can be replaced with advanced shot-detection or segmentation logic.

ii) Metadata stored with embeddings can be extended to track file paths, timestamps, etc.

iii) For production, swap FAISS for a proper vector database such as VDMS, Milvus, or Pinecone.

KEY DETAILS

1. Extracts frames and scene descriptions from videos using open-source models (ViT-GPT2 for captions).

2. Computes text embeddings with SentenceTransformers and image embeddings with OpenCLIP.

3. Stores embeddings in FAISS indexes with associated metadata.

4. Retrieves relevant scenes or frames for a text query.

5. Lightweight Python prototype; can be extended to production vector DBs like VDMS, Milvus, or Pinecone.